

ASIKKALA PULKKILANHARJU, Finland

WMO: 027270

Lat: 61.2653N

Lon: 25.5203E

Elev: 249

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -13.1	(c) -7.1	(d) -18.2	(e) 2.1	(f) -12.9	(g) -11.4	(h) 3.0	(i) -7.1	(j) 29.8	(k) 25.3	(l) 27.5	(m) 24.5	(n) 3.2	(o) 330	(p) 0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.0	(c) 78.2	(d) 66.4	(e) 75.0	(f) 64.5	(g) 72.0	(h) 63.0	(i) 68.7	(j) 75.1	(k) 66.7	(l) 73.0	(m) 64.4	(n) 70.8	(o) 7.3	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.2	97.5	72.7	64.2	90.6	71.1	61.7	82.9	67.8	33.0	75.1	31.5	72.8	29.7	71.1	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.0	(b) 21.6	(c) 18.8	(e) -14.9	(f) 80.9	(g) 11.0	(h) 3.7	(i) -22.8	(j) 83.6	(k) -29.2	(l) 85.7	(m) -35.4	(n) 87.8	(o) -43.4	(p) 90.5		
(a) 25.0	(b) 21.6	(c) 18.8	(e) -14.7	(f) 69.7	(g) 11.4	(h) 2.7	(i) -22.8	(j) 71.7	(k) -29.5	(l) 73.2	(m) -35.8	(n) 74.7	(o) -44.1	(p) 76.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	18.6	21.5	27.5	37.7	50.8	58.2	64.1	61.2	53.1	42.2	34.8	27.5	
	DBStd	17.20	12.56	12.66	8.71	6.21	7.22	5.96	5.81	4.75	5.26	5.76	6.99	9.90	
	HDD50	4357	973	797	699	371	79	6	0	0	31	248	455	698	
	HDD65	8654	1438	1217	1164	819	442	216	90	136	356	708	905	1163	
	CDD50	1272	0	0	0	2	103	253	436	348	125	5	0	0	
	CDD65	94	0	0	0	0	1	13	61	19	0	0	0	0	
	CDH74	357	0	0	0	0	7	46	253	51	0	0	0	0	
	CDH80	30	0	0	0	0	0	2	25	3	0	0	0	0	
Wind	WSAvg	7.8	7.9	8.3	8.5	6.9	7.1	7.6	6.9	6.4	7.5	8.4	9.2	9.7	
Precipitation	PrecAvg	23.6	1.9	1.3	1.2	1.1	1.6	2.5	3.3	2.6	1.8	2.6	2.0	1.8	
	PrecMax	28.7	3.6	2.8	1.9	2.2	3.3	4.0	6.6	4.7	3.8	6.0	4.4	3.2	
	PrecMin	20.7	0.5	0.1	0.2	0.2	0.4	1.0	0.6	0.3	0.6	0.5	0.2	0.4	
	PrecStd	2.0	0.7	0.7	0.5	0.6	0.7	0.9	1.6	1.2	0.9	1.3	1.1	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	42.2	47.8	62.4	74.8	78.6	82.8	79.7	68.1	56.5	50.8	46.9	
		MCWB	37.4	37.6	40.2	49.4	58.6	63.6	67.5	67.3	61.3	53.2	48.4	45.5	
	2%	DB	36.9	38.6	42.9	55.6	69.8	74.8	79.8	74.9	64.7	53.8	48.1	41.7	
		MCWB	35.7	36.4	37.6	45.7	57.1	62.1	67.9	64.7	59.2	51.3	46.7	40.1	
	5%	DB	35.1	36.1	39.7	50.8	66.2	71.4	76.9	71.8	62.4	52.3	45.5	39.5	
		MCWB	34.2	34.7	35.7	42.8	55.6	60.5	66.7	62.7	57.8	50.1	44.3	38.2	
	10%	DB	33.3	34.4	37.3	46.6	62.5	68.0	73.7	69.1	60.5	50.4	43.3	37.5	
		MCWB	32.6	33.5	34.8	40.5	52.8	59.1	65.3	61.5	56.8	48.5	42.3	36.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.1	38.3	41.3	50.2	61.6	68.2	70.9	70.0	62.4	54.0	48.6	45.7
			MCDB	38.7	40.4	45.1	59.6	67.9	73.0	78.1	76.4	65.7	55.7	50.2	46.9
2%		WB	36.0	36.5	38.4	46.8	59.2	64.2	69.4	66.7	60.5	52.2	46.8	40.5	
		MCDB	37.0	38.1	41.8	53.6	66.8	71.5	76.2	72.5	63.2	53.4	48.2	41.4	
5%		WB	34.2	35.0	36.7	43.7	57.0	62.0	68.1	64.4	59.0	50.7	44.6	38.4	
		MCDB	34.9	36.2	39.1	49.7	64.0	68.9	74.6	69.6	61.7	51.9	45.6	39.4	
10%		WB	32.8	33.6	35.0	41.2	54.5	60.2	66.5	62.6	57.3	48.6	42.5	36.3	
		MCDB	33.4	34.3	36.9	45.4	61.7	66.5	72.9	67.1	59.7	50.1	43.2	37.2	
Mean Daily Temperature Range	5% DB	MDBR	9.9	9.6	12.0	13.0	13.7	11.9	11.0	10.4	8.7	7.0	5.5	6.6	
		MCDBR	6.0	7.7	12.9	20.7	19.5	16.9	14.4	14.0	11.4	8.8	7.5	7.3	
		MCWBR	5.9	6.9	9.2	11.5	9.9	8.3	7.2	7.0	7.0	7.4	7.1	7.3	
	5% WB	MCDBR	6.1	7.3	10.9	18.1	17.0	14.7	13.2	12.2	10.4	8.7	7.1	7.4	
		MCWBR	6.0	6.7	8.4	11.5	9.8	8.1	7.2	7.2	7.0	7.9	6.9	7.2	
Clear-Sky Solar Irradiance	taub		0.255	0.298	0.325	0.363	0.360	0.356	0.379	0.366	0.341	0.323	0.280	0.364	
	taud		2.082	2.152	2.229	2.342	2.419	2.459	2.410	2.439	2.500	2.404	2.177	2.595	
	Ebn at Noon		143	210	249	263	274	277	266	260	247	206	134	96	
	Edh at Noon		12	23	30	32	32	32	32	29	23	18	11	8	
All-Sky Solar Radiation	RadAvg		73	252	719	1179	1593	1668	1615	1250	736	314	85	40	
	RadStd		14	57	91	123	177	130	160	142	97	56	16	5	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	-263	N/A	N/A	N/A

Nomenclature: See separate page